Walkthrough: Registered Users & Role/Permissions Management

Summary of Completed Work

We built and integrated a complete, database-backed **Registered Users & Role/Permissions Management System** for **INEZA RESTO & COFFEE SHOP**.

1. Database Schema Migration ()

- Added automatic schema verification and migrations for the MySQL `users` table:
  - `permissions` (`TEXT DEFAULT NULL`): stores assigned module capabilities.
  - `loyalty_points` (`INT DEFAULT 0`): persists real customer loyalty points in the database.
  - `role` (`ENUM('customer','staff','admin','chef','waiter','cashier')`): maps granular staff responsibilities.

2. Dedicated Users Administration API ()

Implemented RESTful actions:

- `action=list`: Returns all registered users with search terms and role filtering. Automatically categorizes records into `staff` and `customers`.
- `action=create_staff`: Creates staff credentials with bcrypt password hashing and assigned permissions.
- `action=update_role`: Modifies user roles and permission sets.
- `action=toggle_status`: Instantly toggles accounts between `Active` (`1`) and `Suspended/Deactivated` (`0`).
- `action=grant_points`: Credits or debits customer loyalty balances directly in MySQL.
- `action=delete`: Deletes staff accounts with protection for the primary administrator (`id = 1`).

3. Admin Portal User Interface ()

- **Live Search & Role Filter Bar**:
  - Instant live filter by customer/staff name, email, or Rwandan phone number (`078...`).
  - Dropdown filter by role (`All`, `Staff Members Only`, `Customers Only`, `Administrators`, `Kitchen Chefs`, `Waiters`, `Cashiers`).
- **Authorized Staff Table**:
  - Displays staff ID/email, contact phone, color-coded role badge, module permissions, live status toggle, and action buttons.
- **Registered Customers Table**:
  - Displays real registered customer accounts with user ID (`#CUST-XX`), phone number, email address, loyalty points balance, joined date, and status toggle.

4. Role Gatekeeping & Permission Enforcement ()

- `enforceRolePermissions()`:
  - Evaluates the logged-in staff member's role upon login.
  - **Super Admin (`admin`)**: Full access across all modules.
  - **Head Chef (`chef`)**: Restricts access to settings, financial analytics, promos, and user management; keeps Kitchen KDS and Live Orders active.
  - **Floor Waiter (`waiter`)**: Restricts access to backend reports, kitchen settings, and user management; keeps Table Dining Tracker and Orders active.
  - **Cashier (`cashier`)**: Restricts table floor layout and system settings; keeps POS Orders and Receipts active.
- **Interactive Toggles & Quick Grant**:
  - Account status toggle button updates the database with instant feedback.
  - "+ Grant Pts" modal credits customer points directly to MySQL.

Verification Results

Endpoint / Action	Input Payload	Result	Status
`GET api/users.php?action=list`	None	Returned 10 registered accounts (1 staff, 9 customers)	✅ Passed
`POST api/users.php?action=grant_points`	`id: 9, points: 1500`	Granted 1500 pts to Patrick Kayonga (Balance: 1500 Pts)	✅ Passed
`POST api/users.php?action=toggle_status`	`id: 9`	Toggled account to suspended (`is_active: 0`)	✅ Passed
`POST api/users.php?action=toggle_status`	`id: 9`	Toggled account back to active (`is_active: 1`)	✅ Passed
Search Filter	Filter by `Patrick` / `0788...`	Accurately displayed matching customer account	✅ Passed